

GRB 090809

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ABSTRACT

We present the complete timing and time-resolved spectral analysis of the single-pulse burst GRB 090809 (*Fermi*/GBM trigger bn090809978), the eighth burst of a 106-burst single-pulse campaign run on a staged, gate-reviewed pipeline. The burst is a clean FRED ($\varphi = 0.199 \pm 0.013$, $R^2 = 0.98$) with $T_{90} = 9.78^{+0.31}_{-0.19}$ s, and it carries the campaign’s largest spectral-lag fraction: the window-scanned lag is $\tau = 0.86 \pm 0.05 \pm 0.07_{\text{win}}$ s at 52σ , giving $\tau/T_{90} = 8.8\%$ — edging past burst #1’s 8.5% at a comparable duration. The wavelet variability estimator returns 215 ± 17 ms — the *fifth* burst on that grid rung, now the modal value of the mapped ladder (181/215/256 ms) — while the canonical windowed method finds 24.9 ± 2.7 ms and Haar gives 1145 ± 28 ms, an intra-burst estimator spread of a factor 46. Nine-block spectroscopy shows a fifth consecutive epoch-dependent synchrotron-line-of-death pattern: the uniform Band index sits above $-2/3$ through the first five blocks ($\alpha = -0.23$ to -0.55) and crosses in the decay ($-0.82 \rightarrow -0.93$), with E_p decaying $445 \rightarrow 73$ keV across the reliable blocks. No blackbody-bearing model wins any bin at the AIC level, though the nested likelihood-ratio gate finds significant, edge-clean blackbodies in three mid-burst blocks ($kT = 29, 25, 19$ keV) — a metric split we report explicitly, sitting between the BB-loaded bursts #4/#7 and the winner-level thermal absences. The integrated spectrum is a single-break SBPL ($E_b = 99 \pm 15$ keV, $\beta = -2.13 \pm 0.08$) in a $\Delta\text{AIC} = 0.2$ tie with Band+cutoff. Three raw block winners are bound-railed and disclosed invalid; the 24×10 grid renders under the no-model-dropped rule with two labeled engine-refusal cells; b_0 ’s effective-area factor rails at 0.80 and is disclosed.

Keywords: Gamma-ray bursts (629) — Time domain astronomy (2109)

1. INTRODUCTION

GRB 090809 triggered the *Fermi* Gamma-ray Burst Monitor (Meegan et al. 2009) on 2009 August 9. As burst #8 of the single-pulse campaign it runs on the pipeline frozen from bursts #1–#2. Its contributions to the cross-burst picture are extremal rather than novel: the largest lag fraction, the fifth occupation of the wavelet estimator’s 215 ms rung (establishing it as the ladder’s modal value), a fifth consecutive epoch-resolved line-of-death crossing, and the campaign’s sharpest thermal metric split: no BB-bearing winner in any of nine blocks, yet three significant nested-test blackbodies.

2. OBSERVATIONS, DETECTORS, AND APPROVED SELECTIONS

We use TTE data from NaI detectors n_3 , n_4 , n_5 and BGO b_0 (Figure 1), with the spectral convention of Chand et al. (2020): NaI 8–900 keV excluding the 30–40 keV K-edge, BGO 0.2–38 MeV. Background and source selections are the human-approved Stage-1 decisions (source window $(-0.30, 15.50)$ s; Figures 2–3). Effective-area factors float in $[0.8, 1.2]$; the integrated

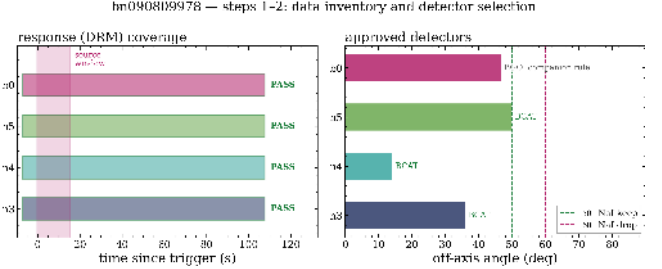


Figure 1. Detector selection for GRB 090809: source angles with the 50° rule; selected detectors n_3 , n_4 , n_5 , b_0 .

winner carries $n_4 = 1.05$, $n_5 = 1.05$, and b_0 railed at 0.80 — the recurrent BGO-side rail, disclosed as a standing caveat.

3. TIMING ANALYSIS

All timing runs on the approved selections, reference NaI n_4 .

3.1. Durations

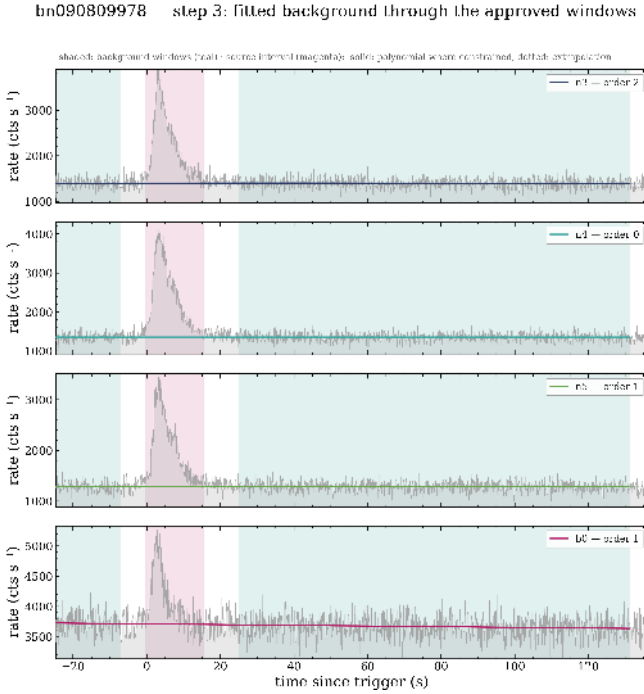


Figure 2. Approved background intervals and per-detector polynomial fits.

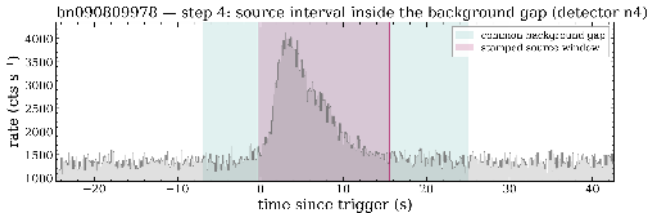


Figure 3. Approved source interval over the background-subtracted light curve.

The windowed duration is $T_{90} = 9.78^{+0.31}_{-0.19}$ s spanning [1.48, 11.26] s (covariance-true Monte Carlo; window not truncated), with $T_{50} = 4.12$ s ($T_{50}/T_{90} = 0.42$).

3.2. Pulse Shape

The Kocevski et al. (2003) profile wins the shape comparison, and the two-sigmoid of Gowri et al. (2025) gives $\varphi = s_l/s_r = 0.199 \pm 0.013$ ($R^2 = 0.98$): a clean FRED (Figure 4). The campaign’s shape axis now holds six FREDs ($\varphi = 0.017$ –0.32; the count includes the unpapered burst #3, $\varphi = 0.13$) and two mixed pulses ($\varphi \approx 0.40$).

3.3. Minimum Variability Timescale

All three estimators return measurements (Figure 5): the windowed framework of Bala et al. (2026) — the canonical value — finds 24.9 ± 2.7 ms ($\Delta t = 2$ s segment, detection); the global CWT (Vianello et al. 2018) returns 215 ± 17 ms — the *fifth* burst on this rung (after

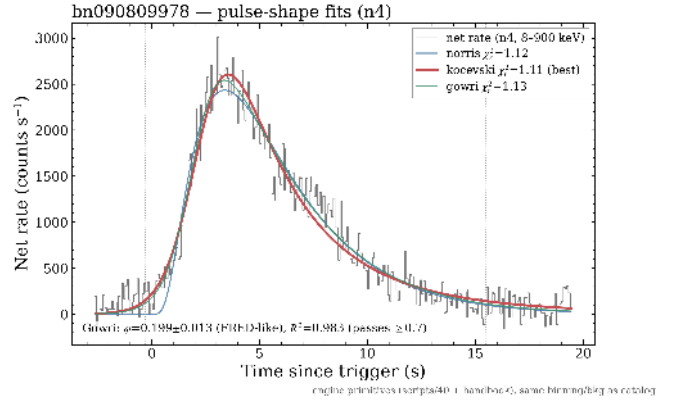


Figure 4. Pulse-shape fits to the net light curve; Kocevski wins; $\varphi = 0.199 \pm 0.013$, a clean FRED.

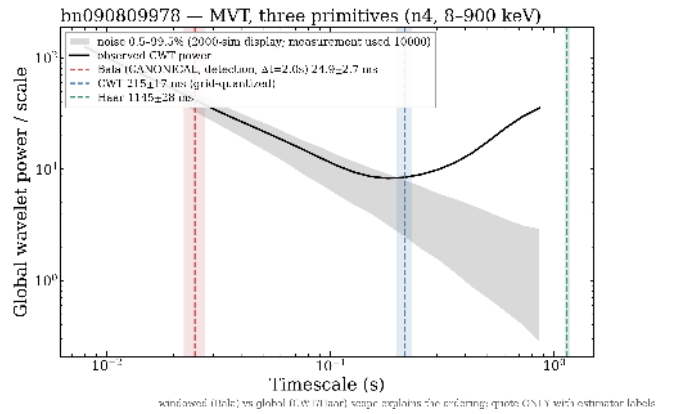


Figure 5. Minimum-variability estimators with explicit labels: canonical windowed 24.9 ± 2.7 ms; CWT 215 ± 17 ms (fifth occupation of the modal rung); Haar 1145 ± 28 ms.

#1, #2, #3, #5; the count includes the unpapered burst #3, whose products carry the identical value), making 215 ms the modal value of the estimator’s mapped ladder (181/215/256 ms) and putting the grid-quantization finding beyond doubt; the Haar cross-check (Golkhou & Butler 2014) gives 1145 ± 28 ms, the campaign’s slowest Haar value and an intra-burst estimator spread of a factor 46.

3.4. Spectral Lag: the Largest Fraction

Scanning the cross-correlation fit window around the peak (half-widths 1.0–4.0 s; Figure 6) gives $\tau = 0.86^{+0.05}_{-0.05} \pm 0.07_{\text{win}}$ s (25–50 keV vs. 100–300 keV, soft lagging hard; detector n_4 , 16 ms resolution, peak significance 52σ); the scan members run 0.75–0.88 s. The lag fraction $\tau/T_{90} = 8.8\%$ is the campaign’s largest, marginally above burst #1’s 8.5% at a comparable duration — the lag-width plane now spans zero to $\sim 9\%$ with no monotonic trend in duration.

3.5. Per-band Durations

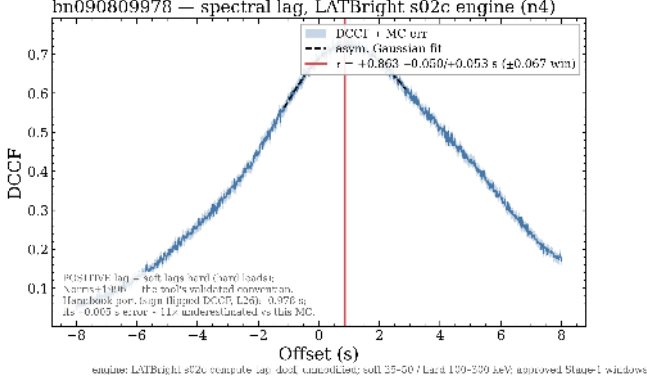


Figure 6. Cross-correlation lag with the window-scan systematic; the quoted value is the pulse-scaled member closest to the scan median.

Per-band light curves and the $T_{90}(E)$ relation are shown in Figure 7; the population slope of Qin et al. (2013) is drawn for reference only.

4. SPECTRAL BINNING

Nine Bayesian blocks plus the integrated spectrum (Figure 8), from the frozen 3ML Bayesian-block + significance hybrid (blocks computed 2026-08-12, reused unchanged).

5. TIME-RESOLVED AND TIME-INTEGRATED SPECTROSCOPY

5.1. Method

All 24 campaign models fit to every bin (PGstat; AIC ranking (Akaike 1974)); no LLE/LAT data exist for this burst (inclusion gated on data availability, never detection significance). The 24×10 grid renders under the no-model-dropped rule with two labeled engine-refusal cells (Band+CPL in the integrated spectrum and in block 8); per-bin all-model parameter tables accompany the products.

5.2. Integrated Spectrum

The integrated spectrum (0.88, 15.18) s is won by the single-break SBPL (Figure 9): $\alpha = -0.86 \pm 0.05$, $E_b = 99 \pm 15$ keV, $\beta = -2.13 \pm 0.08$ (AIC = 5885.8), with Band+cutoff at $\Delta\text{AIC} = 0.2$ — another statistical tie at the top of the ranking. The uniform-Band integrated $\alpha = -0.83$ sits below the line of death, as expected for a burst whose violation is confined to its early epochs.

5.3. Time-Resolved Story

In the uniform Band convention α runs $-0.23, -0.44, -0.34, -0.55, -0.55$ across blocks 0–4 — all above the $-2/3$ line of death — then crosses in the decay: $-0.82, -0.86, -0.93$ in blocks 5–7. Block 8’s

uniform-Band read ($\alpha = -1.73$, $E_p = 201$ keV) comes from a β -railed fit ($\beta \rightarrow -5$, the bound) in the burst’s faint tail and is excluded from the trend on that ground. This is the fifth consecutive burst in which the synchrotron violation is an epoch property — the campaign has yet to see a single-pulse burst that is compliant or violating *throughout*. E_p decays $445 \rightarrow 73$ keV across the reliable blocks (Figure 10), a factor of 6.

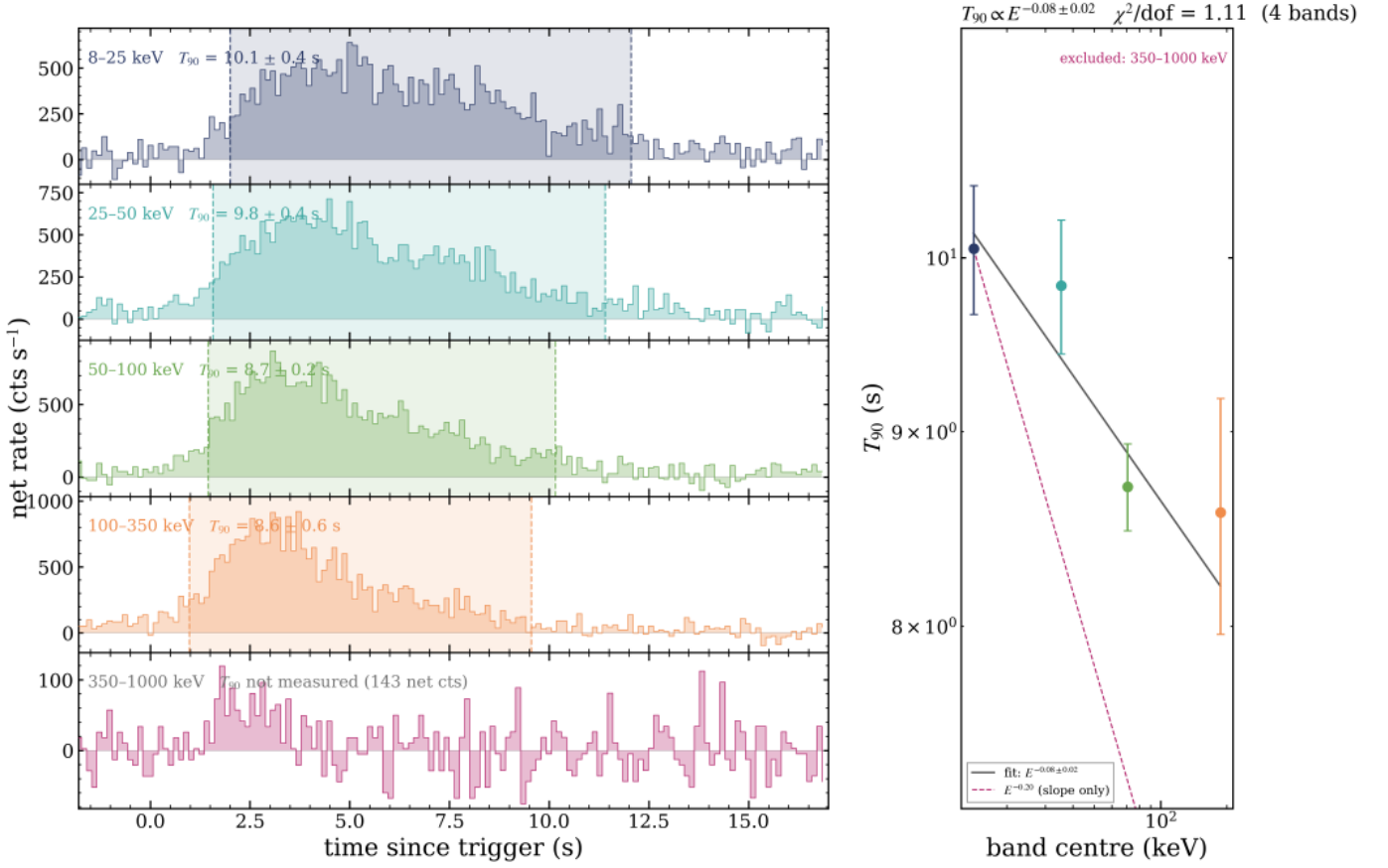
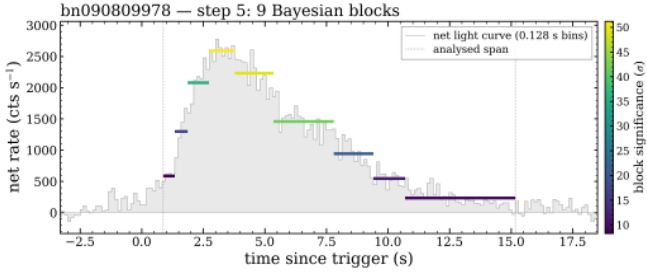
Model preference is again tie-dominated: the winner margins in all ten spectra are $\Delta\text{AIC} < 1.8$, and three raw block winners (blocks 1, 3, 8) are bound-railed and disclosed *invalid* — in block 1 the invalid winner ties at $\Delta\text{AIC} = 0.00$ with a DSBPL twin minimum that is itself flagged invalid — an invalid–invalid degeneracy we flag for the engine-hardening list. No blackbody-bearing model wins any bin at the AIC level (Figure 11 shows the representative block-5 winner) — but the step-9 nested-LRT gate (Figure 12) finds significant blackbodies in blocks 2–4 ($kT = 28.7, 25.0, 19.2$ keV, $\text{LRT} \geq 9.2$, Wien peaks 112/98/75 keV, all edge-clean). The two metrics disagree here more sharply than in any earlier burst, and we report both rather than adjudicate: at winner level the burst is thermal-free; at nested-test level it carries a three-block ~ 20 –29 keV track.

6. QUALITY CONTROL AND PROVENANCE

Step-9 QC (Figure 12) is built from the canonical campaign fit table; every figure carries its producing script’s SHA-256 in a JSON sidecar; numbers are recomputed from product files; all figures pass a fresh-context vision gate before delivery, enforced by a pre-delivery hook. Fits come from the campaign’s one-invocation 24-model engine run with best-minimum-per-(bin, model) canonicalization.

7. SUMMARY

1. GRB 090809 is a clean FRED ($\varphi = 0.199 \pm 0.013$) with $T_{90} = 9.78^{+0.31}_{-0.19}$ s.
2. Its lag fraction is the campaign’s largest ($\tau = 0.86 \pm 0.05 \pm 0.07_{\text{win}}$ s; $\tau/T_{90} = 8.8\%$) — the lag-width plane now spans zero to $\sim 9\%$.
3. The CWT estimator returns its modal rung for the fifth time (215 ± 17 ms); the canonical MVT is 24.9 ± 2.7 ms; the intra-burst estimator spread reaches a factor 46.
4. Nine-block spectroscopy gives the fifth consecutive epoch-dependent line-of-death pattern: violation through blocks 0–4 (α up to -0.23), crossing in the decay; E_p decays $445 \rightarrow 73$ keV.

bn090809978 — step 7: energy-resolved T_{90} (detector n4); dashed = t_5 , t_{95} **Figure 7.** Per-band light curves with individual T_{90} spans and the $T_{90}(E)$ relation.**Figure 8.** Bayesian-block spectral bins over the light curve.

5. No blackbody-bearing model wins any bin at the AIC level, while the nested-LRT gate finds

three significant edge-clean blackbodies (blocks 2–4, $kT = 29\text{--}19$ keV) — the campaign’s sharpest metric-dependence case for thermal claims.

6. The integrated spectrum is a single-break SBPL ($E_b = 99 \pm 15$ keV) in a $\Delta\text{AIC} = 0.2$ tie with Band+cutoff.
7. Standing caveats: b_0 rails at 0.80; three raw block winners bound-railed invalid (block 1’s ties a valid twin at $\Delta\text{AIC} = 0.00$); two labeled engine-refusal cells; all numbers provisional pending the PI’s campaign review.

APPENDIX

A. ALL-MODEL MONTAGES

Complete 24-model montages for the integrated spectrum and all nine blocks; the winner panel is marked in each; engine-refusal cells are labeled.

REFERENCES

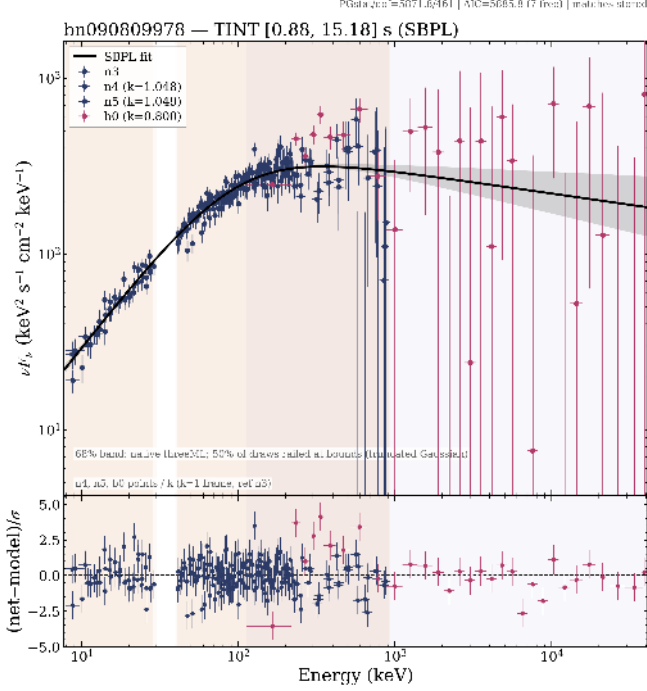


Figure 9. Integrated-spectrum winner: SBPL with a single break at 99 keV. 68% band from propagated posterior draws; PGstat/dof, AIC, and EACs stamped on-figure.

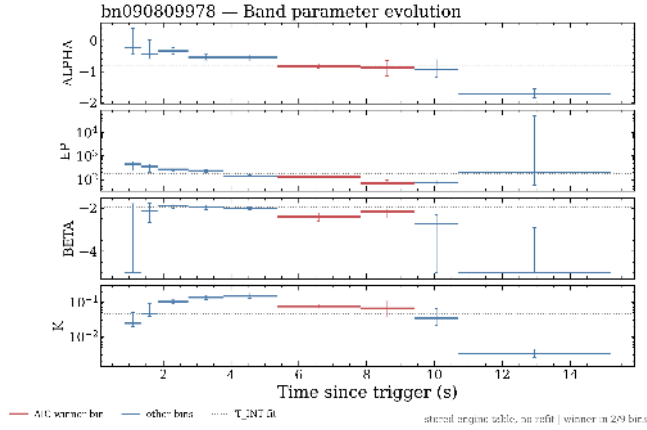


Figure 10. Band-family parameter evolution: the line-of-death crossing between blocks 4 and 5 and the monotonic E_p decay.

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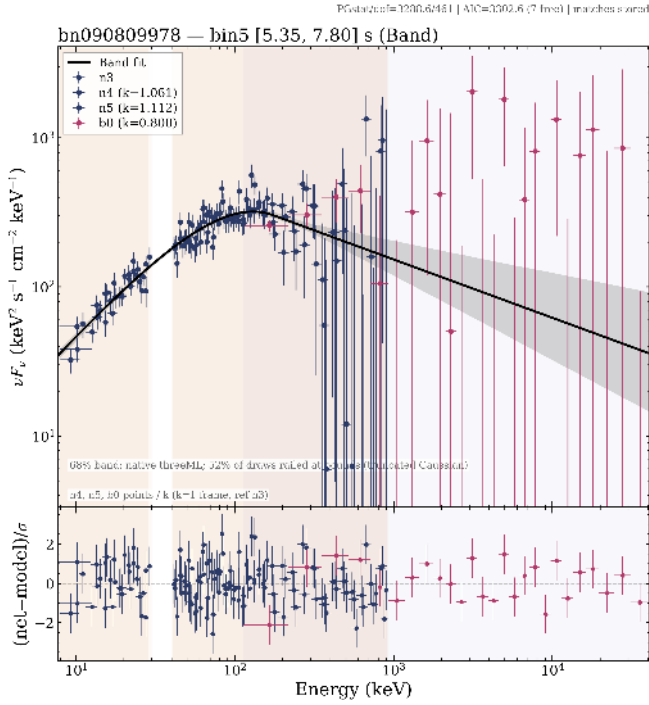


Figure 11. Block 5 (5.35, 7.80) s: Band winner at the line-of-death crossing ($\alpha = -0.82$).

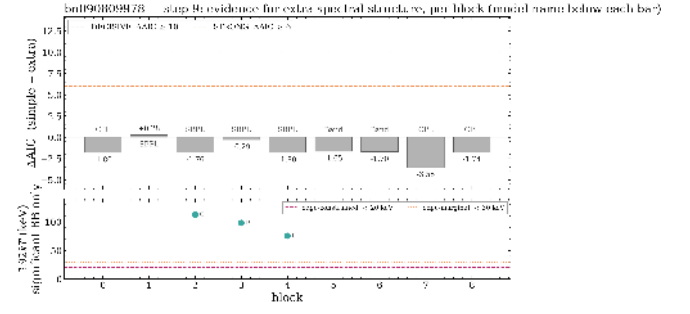


Figure 12. Step-9 scorecard: per-block evidence for extra spectral structure (top) and the 3.92 kT Wien-peak edge check (bottom).

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doi: [10.3847/1538-4357/aad6ea](https://doi.org/10.3847/1538-4357/aad6ea)

bn090809978 — TINT [0.88, 15.18] s — 24 models, AIC-ordered — winner SBPL

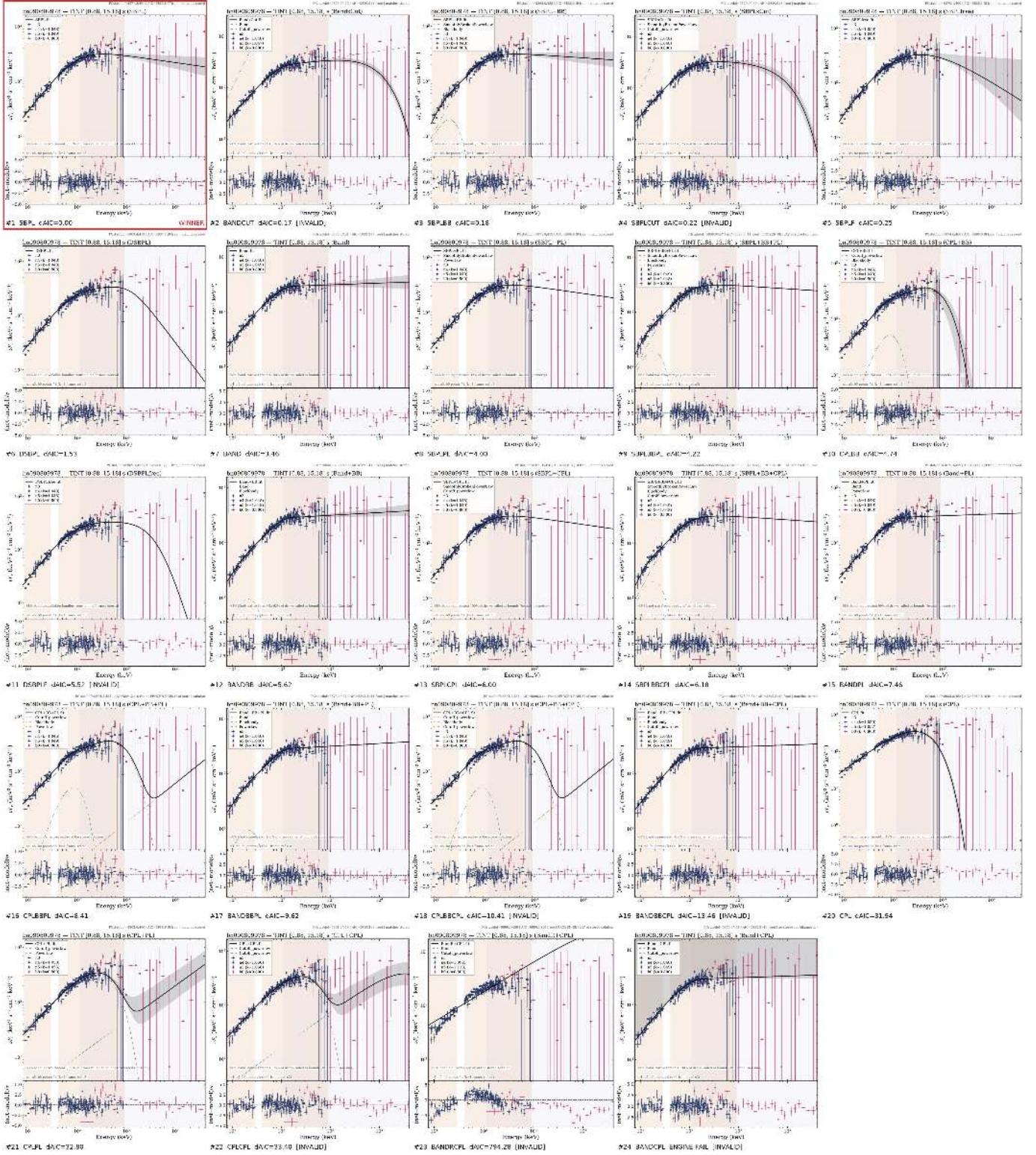


Figure 13. Integrated spectrum: all 24 models (Band+CPL: labeled refusal cell).

bn090809978 — bin0 [0.88, 1.34] s — 24 models, AIC-ordered — winner CPL

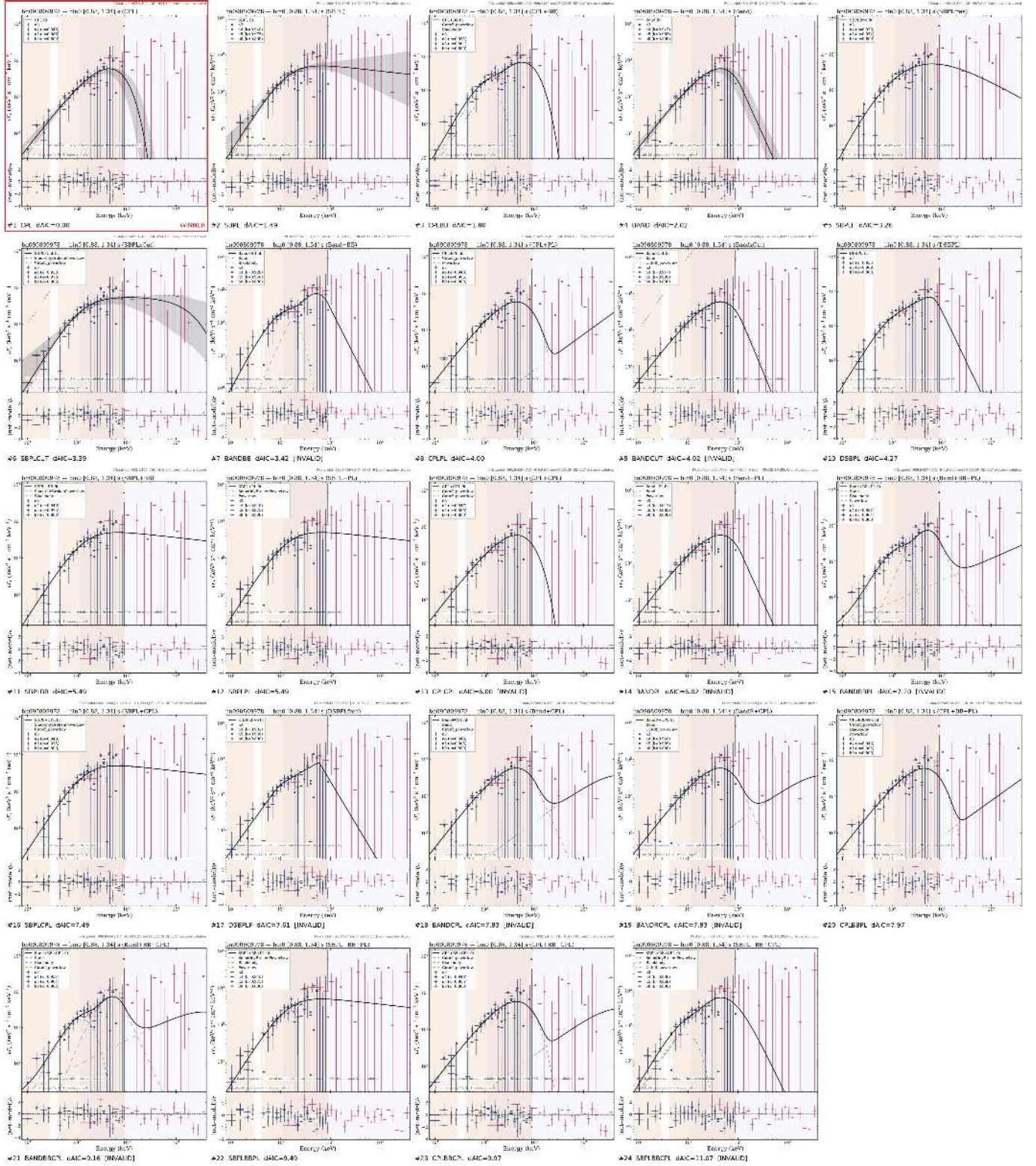


Figure 14. Block 0: all 24 models.

bn090809978 — bin1 [1.34, 1.87] s — 24 models, AIC-ordered — winner SBPLCUT

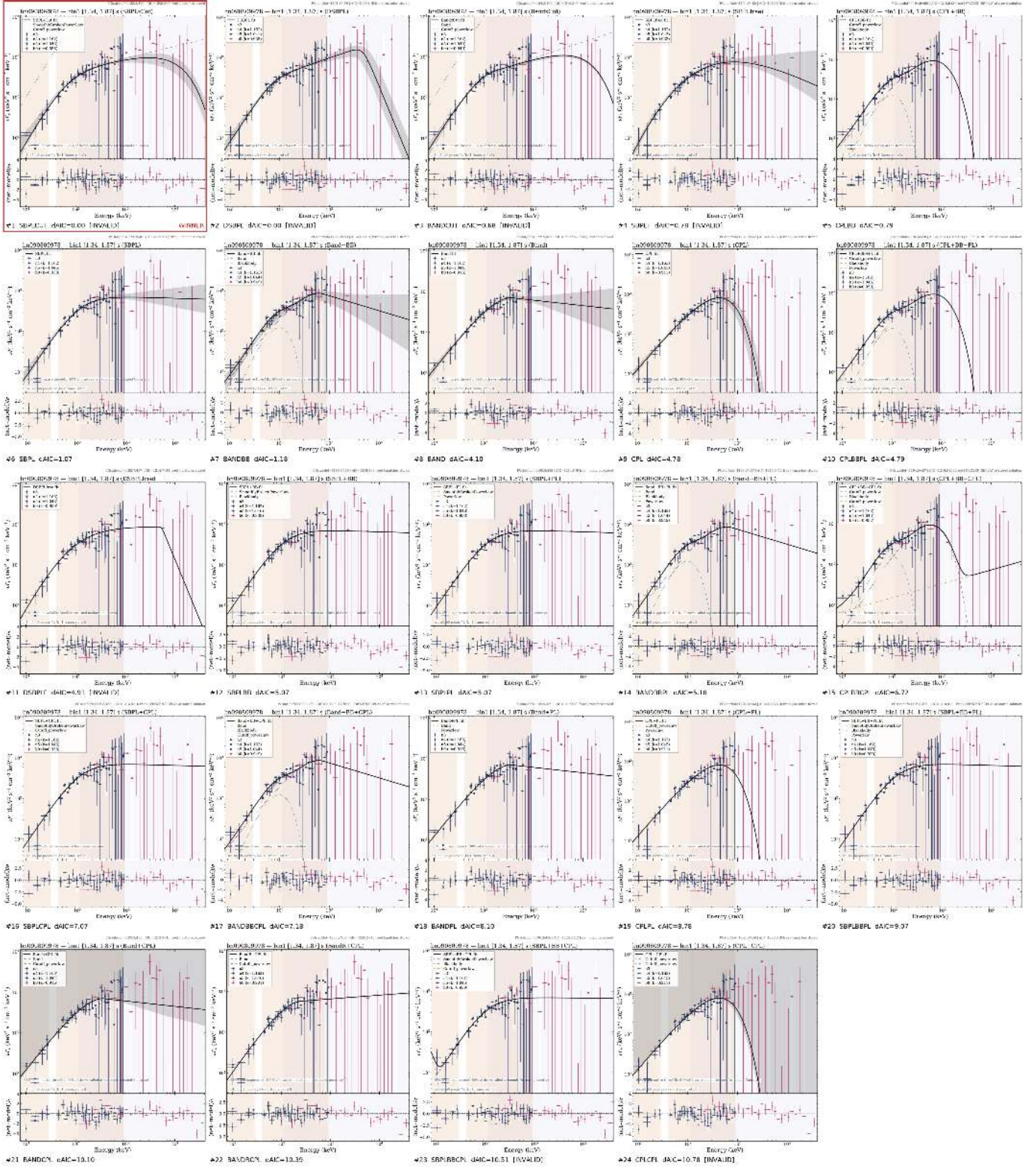


Figure 15. Block 1: all 24 models.

bn090809978 — bin2 [1.87, 2.74] s — 24 models, AIC-ordered — winner SBPL

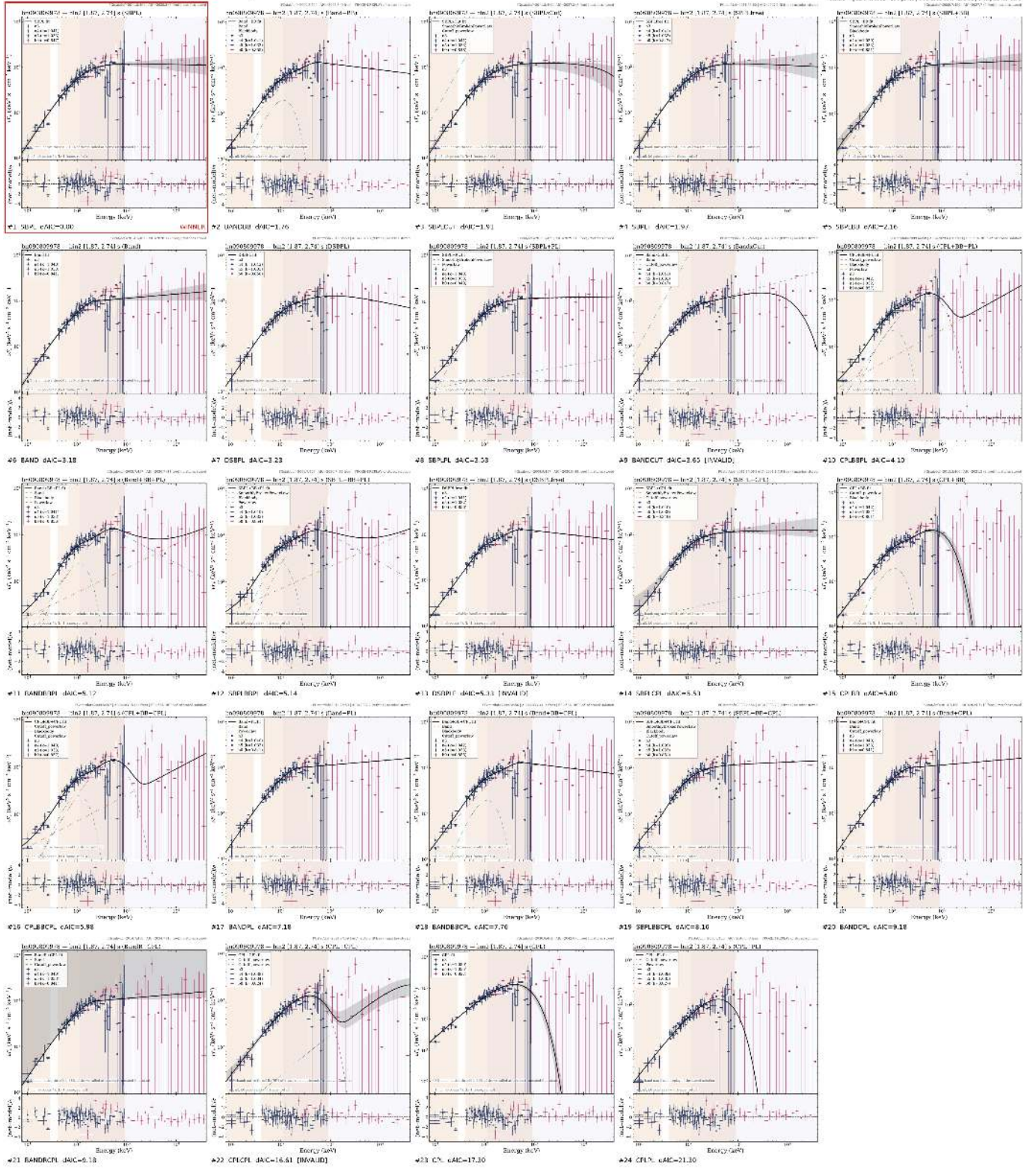


Figure 16. Block 2: all 24 models.

bn090809978 — bin3 [2.74, 3.78] s — 24 models, AIC-ordered — winner SBPLF



Figure 17. Block 3: all 24 models.

bn090809978 — bin4 [3.78, 5.35] s — 24 models, AIC-ordered — winner SBPL



Figure 18. Block 4: all 24 models.

bn090809978 — bin5 [5.35, 7.80] s — 24 models, AIC-ordered — winner BAND



Figure 19. Block 5: all 24 models.

bn090809978 — bin6 [7.80, 9.41] s — 24 models, AIC-ordered — winner BAND

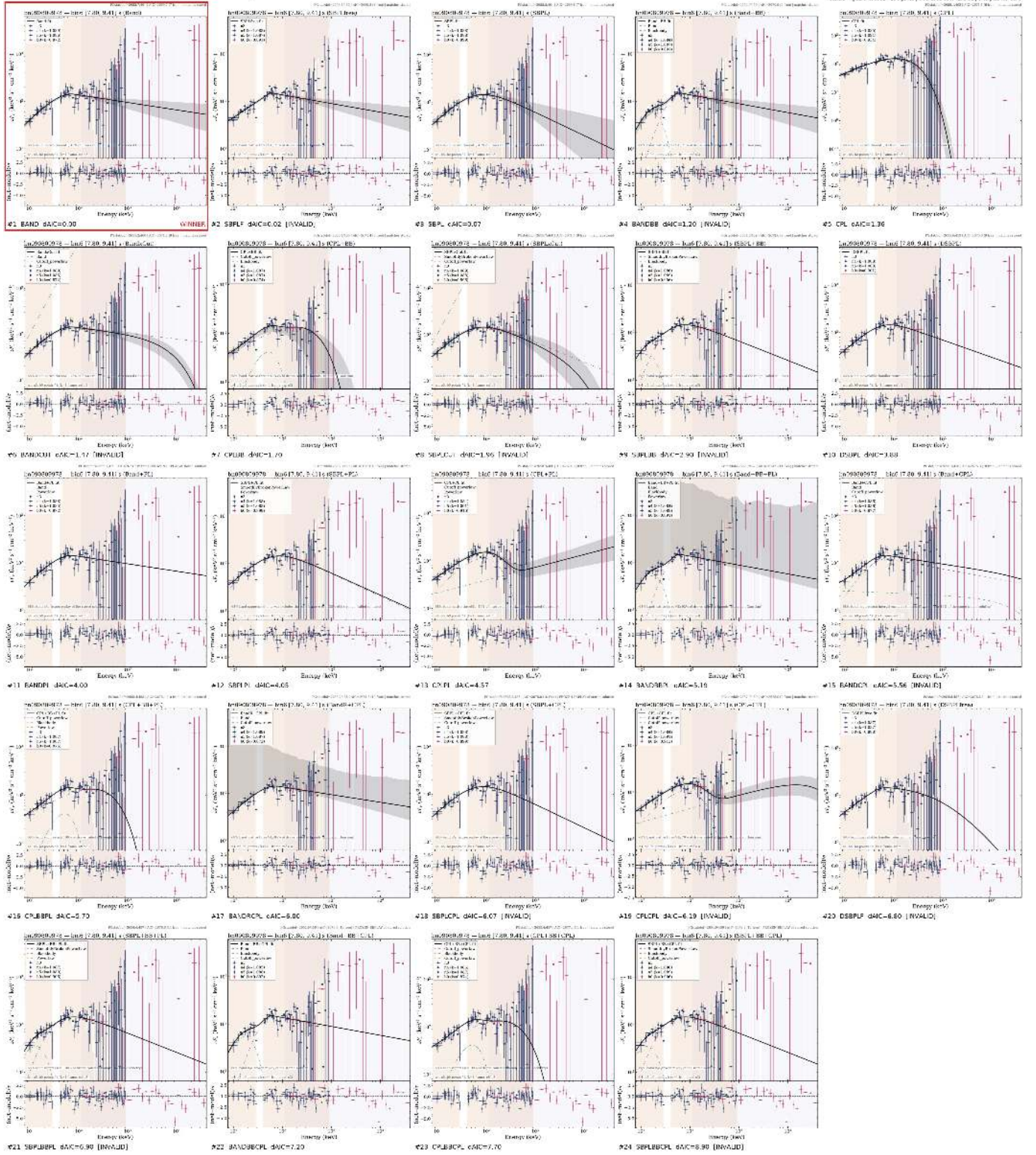


Figure 20. Block 6: all 24 models.

bn090809978 — bin7 [9.41, 10.70] s — 24 models, AIC-ordered — winner CPL



Figure 21. Block 7: all 24 models.

bn090809978 — bin8 [10.70, 15.18] s — 24 models, AIC-ordered — winner SBPL



Figure 22. Block 8: all 24 models (Band+CPL: labeled refusal cell).